Rakesh Lakshmanan

Software Engineer | Backend · Distributed Systems · Full-Stack

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Professional Summary

Software engineer with ~3 years at **Zoho Corporation, a global SaaS company**, building **highly available, fault-tolerant distributed systems in Java**: a platform framework adopted by **20+ product teams**, an HA/DR topology sustaining **99.95% uptime**, and production deployments across **100+ nodes**. Recent **MSc Computer Science (Distinction)** graduate from Trinity College Dublin, specialising in Future Networked Systems. Hands-on across **Java 17** / **Spring Boot, Python** / **FastAPI, and React** / **TypeScript**, building microservices, event-driven pipelines, and cloud-native deployments on Kubernetes with CI/CD and observability. Seeking **backend, distributed systems, or full-stack** engineering roles.

Education

Trinity College Dublin 09/2025 – 08/2026
MSc in Computer Science – Future Networked Systems – Grade 1:1 (Distinction)
Relevant Coursework – Distributed Systems, Advanced Software Engineering, AI, ML
Dissertation – *Modern Delay-Tolerant Email*: email and anti-spam tooling for high-delay cislunar networks, supervised by Stephen Farrell (co-author of RFC 9174)

St. Joseph's College of Engineering, India 06/2019 – 04/2023
Bachelor's in Computer Science – Grade 1:1 (92.3%)

Technical Skills

Languages: Java (JDK 17), Python, TypeScript, JavaScript, SQL, C
Backend & APIs: Spring Boot, Spring Security, FastAPI, Node.js, RESTful APIs, Microservices, Event-Driven Architecture, Multithreading & Concurrency
Frontend: React, TypeScript, HTML, CSS
Distributed Systems: High Availability & Disaster Recovery, Fault Tolerance, Replication, Scalability, Saga Pattern, Distributed Locking, Reconciliation
Databases & Messaging: PostgreSQL (PostGIS), MySQL, MSSQL, ClickHouse, SQLite, MongoDB, Redis, RabbitMQ
Cloud & DevOps: AWS (EC2, RDS), GCP (GKE), Docker, Kubernetes, Helm, Terraform, Istio, GitHub Actions CI/CD, Prometheus, Grafana
Networking: TCP/IP, DNS, HTTP, Delay-Tolerant Networking (DTN)
AI & Tools: AI Coding Assistants (GitHub Copilot, Claude Code, Cursor), LLM Integration (Gemini), scikit-learn, Git, Gradle, Maven, Linux

Experience

Member Technical Staff (Software Engineer) – Zoho Corporation, Chennai, India 06/2023 – 08/2025

- Scaled **Mickey, Zoho's internal Java platform framework**, to **20+ product teams** by architecting reusable components that cut development time and onboarding friction, contributing to **~10% customer growth** across adopting teams.
- Achieved **99.95% uptime** in distributed production environments by designing a **four-node High-Availability and Disaster Recovery (HA/DR) topology** with configuration-driven scheduling and Robocopy-based replication.
- Reduced manual intervention during failovers by **~60%** and recovery time by **~35%** by automating upgrade workflows and hardening server-to-server file transfer in the **failover and recovery mechanisms**.
- Contributed to scalability planning, architecture design, and **performance tuning** of distributed systems supporting production deployments across **100+ nodes**.
- Enabled **10 analytics-heavy teams** to adopt fast columnar reads by integrating **ClickHouse** as an optional database backend alongside PostgreSQL/MSSQL in the framework.
- Designed and developed the **Read-Only Server module framework**, a reusable solution adopted by **15+ product teams**, and supported its integration into their products.
- Built **DBScanner**, a tool relied on regularly by **20+ product teams** that compares two databases and produces clear, structured difference reports to streamline migrations.

- Built **PGMigrate**, turning complex manual PostgreSQL version upgrades into a straightforward, repeatable workflow and cutting upgrade effort by ~**30%**.
- Resolved **200 long-standing defects** in the main codebase by partnering with the **QA team** to prioritise, reproduce, and fix issues through to root cause.
- Mentored two summer interns on HA topology design, fault-tolerance patterns, and production deployment practices.

Project Trainee – Zoho Corporation, Chennai, India 01/2023 – 05/2023

- Reduced manual testing effort by **60%** and expanded regression coverage by **40%** by architecting a **test automation framework** for Java Swing applications.
- Led development of a **Java NIO filesystem provider** that unified **HDFS** data access for **15+ applications**, improving team efficiency by **20%**.
- Cut file-transfer error rates from **22% to 7%** by building **FileSync**, a Java NIO-based file replication utility with robust error handling, integrated into the HA module.

Summer Intern – Zoho Corporation, Chennai, India 04/2022 – 09/2022

- Built a **SQLite-backed queue** in Java guaranteeing FIFO ordering for task and message processing.
- Resolved **500+ production and support tickets** over six months and shipped **100+ bug fixes** to the test automation platform.

Projects

SmartEnoughCity – Cloud-Native Urban Transport Intelligence Platform GitHub

- Built a **4-service microservices platform** ingesting real-time data from **15+ Dublin transport APIs** into PostgreSQL/PostGIS, with **scikit-learn models** powering bike-availability risk, ridership forecasts, and disruption alerts on a React dashboard.
- Provisioned GKE with **Terraform** and a **Helm umbrella chart**, with Istio and **Prometheus/Grafana monitoring**; CI/CD via GitHub Actions (Workload Identity Federation) with a ∼**600-test suite** enforcing a 75% coverage gate on every PR.
- **Tech Stack**: Java 17, Spring Boot, Python, FastAPI, scikit-learn, React, TypeScript, PostgreSQL + PostGIS, Kubernetes (GKE), Helm, Terraform, Istio, GitHub Actions, Prometheus, Grafana

Distributed Traffic Booking System – Multi-Region Microservices Booking Platform GitHub

- Architected a **6-service booking platform** across three geographic data partitions (Ireland, UK, France), using a **Saga orchestrator** with compensating transactions to coordinate cross-region journey bookings.
- Prevented double-booking under **concurrent load** with **Redis distributed locks** per road segment and time slot, backed by a **saga reconciler** that auto-heals partition inconsistencies every 120 seconds.
- **Tech Stack**: Python 3.12, FastAPI, SQLAlchemy (async), PostgreSQL, Redis, RabbitMQ, React, TypeScript, Docker, AWS EC2/RDS

DevVault – Full-Stack AI-Powered Developer Bookmark Manager Live | GitHub

- Built a **production-deployed** bookmark manager that automatically summarises and tags every saved page using **Google Gemini 2.5 Flash**, with JWT auth, Google OAuth2 SSO, and social sharing.
- Engineered an **event-driven AI pipeline** (Spring `@TransactionalEventListener` + dedicated thread pool) that fully decouples AI latency from user-facing response time, with a **reconciler** that reprocesses failed jobs.
- **Tech Stack**: Java 17, Spring Boot, Spring Security, PostgreSQL, Gemini 2.5 Flash, React, TypeScript, Docker, Kubernetes (GKE), GitHub Actions

Certifications & Achievements

- **AI Certifications (2026)**: **Building AI Agents and Agentic Workflows** – IBM (Credential); AI Professional Certificate – Google (Credential); AI Infrastructure and Operations Fundamentals – NVIDIA (Credential); Architecting Scalable Cloud AI Infrastructure – Coursera (Credential)
- Awarded **Mickey Emerging Programmer (Q4 FY23)** at Zoho for high-impact contributions to the internal Java framework.
- Recipient of **LRG Naidu Educational Trust Scholarship (ELGI Scholar)** for undergraduate academic excellence.